

YASHVARDHAN GOEL

+91 9310720186 ◇ Delhi, India (Open to Relocating to Boston)

[Email](#) ◇ [LinkedIn](#) ◇ [GitHub](#) ◇ [Website](#)

SUMMARY

Agentic AI engineer building **production autonomous agents**, not autocompletable-in-an-IDE workflows, but agents with their own **memory**, **MCP-based tools**, and **self-improvement loops**. I ship with the **Agno framework**, design **MCP tool definitions**, and build the feedback loops that let agents improve without supervision. Currently building a **next-generation agent memory system** beyond knowledge graphs, with early benchmarks showing **200–300× lower retrieval latency**, **~50% less token consumption**, and an **83% anticipatory-hit rate**, making agents **proactive** rather than reactive.

EDUCATION

M.Sc. Biological Sciences, Birla Institute of Technology and Science, Pilani (BITS Pilani) 2020-2025

Relevant Coursework: Computer Programming, Probability and Statistics, Mathematics - I, II

B.E. Manufacturing Engineering, Birla Institute of Technology and Science, Pilani (BITS Pilani) 2021-2025

Relevant Coursework: Mathematics - III, Engineering Optimization

SKILLS

Technical Skills Agentic AI, LLMs, Agno, MCP (Model Context Protocol), LangChain, Machine Learning, Natural Language Processing, Deep Learning, Computer Vision, Milvus, Github Actions, OOPs, Docker, Nginx, FastAPI, Git, AWS, Redis, PostgreSQL

EXPERIENCE

Software Developer - AI/ML Nov 2025 - Apr 2026

Mira, Aroha Labs *Singapore, Singapore*

- Built **production-grade AI agents** using the Agno framework with **custom memory architectures** and **Milvus-powered hybrid retrieval** pipelines for improved reasoning and **context-aware** responses
- Developed a **learning pipeline** that enables AI agents to continuously improve through an **experiential feedback loop**, leveraging techniques such as **Natural Language Inference (NLI)** and **Human-in-the-Loop** systems
- Engineered **integrations** with enterprise tools (**Google Workspace**, **M365**, **Hubspot**, etc.) using **MCP-compatible tool definitions**, enabling AI agents to understand workflows and perform **cross-platform actions**
- Architected **microservices** on AWS with **SQS/SNS**, **ECS clusters**, **ALB**, and **VPC-isolated** services
- Set up **CI/CD** pipelines via **GitHub Actions** for **automated deployment** across all **microservices** on AWS

AI Engineer Intern May 2024 - Oct 2025

Mira, Aroha Labs *Singapore, Singapore*

- Spearheaded the AI backend for [Delphi Intelligence's](#) chatbot using **LangChain**, **Milvus**, **hybrid-retrieval**, **advanced memory**, and intelligent **routing** to deliver **superfast, context-aware** responses
- Pioneered [Mira Flows](#), a plug-and-play AI platform with **5,000+ users**, by employing a **microservice arch.**
- Built a [Web3 security agent](#) leveraging GoPlus APIs, utilizing **Model Context Protocol (MCP)** for robust, **efficient communication** with the APIs to **deliver real-time, automated security intelligence**

Machine Learning Intern Jan 2024 - Apr 2024

Rocketium, Mycelium Technologies *Bengaluru, India*

- Developed a **fine-tuned U2-Net** model for **analyzing** digital ads that achieved a **50% increase** in engagement
- Conducted a **comparative analysis** of Azure, AWS Rekognition and Google Vision services for image tagging
- Boosted user engagement by **50%** by optimizing ad strategies with **fine-tuned ChatGPT** and **LlamaIndex**

Machine Learning Intern May 2023 - Jul 2023

Hindustan Unilever Ltd. (HUL) *Assam, India*

- Collaborated with **Deloitte** on ML-driven labor allocation, **saving 1.5 hours** daily and increasing production

- Employed **GBM regressor** for quality checks, **reducing BCT by 20 minutes** and **boosting production by 50%**
- Spearheaded the development of a ML model for **energy sustainability**, aimed at **reducing carbon footprint**

PROJECTS

Agent Memory System

In Progress

Personal Project

- Building a **next-generation memory layer** for AI agents beyond **knowledge-graph** approaches; early benchmarks show **200–300× lower retrieval latency** and **~50% lower token consumption** per interaction
- Achieves an **83% average anticipatory-hit rate**, making agents **proactive**, anticipating and initiating the next action instead of the industry-standard **reactive request-response loop**

Handoff

Apr 2026 - Apr 2026

Personal Project - [Github Link](#)

- Built a lightweight **MCP server** in Python that **persists** AI conversation-turn summaries to a **shared CLAUDE.md** file, keeping **Claude Code** and **Claude Desktop** **synchronized** on the latest project context across sessions
- Extended the server to **support multi-user collaboration** over **HTTPS** transport, allowing **multiple developers** to connect their individual Claude Code agents to a **single shared memory server**
- Designed a **platform-scoped CRUD API** (`log_turn`, `list_entries`, `update_entry`, `delete_entry`) where **each client** can only **write to its own section** while having **full read access** to the **shared history**
- Implemented concurrent file-access safety using **fcntl file locking**, **atomic writes via temp-file replacement**, and **regex-based structured parsing** to guarantee **data integrity** across **simultaneous** multi-agent writes

TL;DAI (Too Long; Did AI)

Nov 2025 - Present

Personal Project - [PlayStore Link](#)

- Built an **Android** app that aggregates AI and tech updates from arXiv, Reddit, GitHub Trending, and HN
- Designed an **AI summarization pipeline** to convert long form articles and papers into concise summaries
- Focused on **privacy** by ensuring **no personal data collection** and **local storage of user preferences**

Computational Discovery of Battery Materials

May 2023 - May 2024

Supervised by - [Prof. Tribeni Roy](#)

- Adapted deep learning frameworks to **predict the chemical compound properties** with **90%** accuracy
- Built a **BERT based text extraction tool** for **data mining** in research articles, achieving **85%** accuracy
- Employed **hierarchical GNNs** to discover **novel** battery materials with better properties for the battery industry

Tool for Mining Biological Literature

Mar 2023 - Dec 2023

Supervised by - [Prof. Rita Sharma](#)

- Extracted numerous research papers featuring **rice-associated genes**, using **NLP** and **PubMed's API**
- Programmed a **fine-tuned BERT** model to classify rice genes by **isolation methods** on a **self-curated** dataset
- Designed a database-driven website to aid **biological researchers** by cataloging genes and their isolation methods

ACHIEVEMENTS

Taiwan Experience Education Program Scholar (TEEP)

Sep 2025 - Nov 2025

Supervised by - [Prof. Wang Chin-Tsan](#)

Yilan County, Taiwan

- TEEP is an initiative of the **Ministry of Education, Government of Taiwan**, providing international researchers with hands-on research exposure and cultural immersion in Taiwan
- The topic of research was **Predicting time-series voltage of MFC's using hybrid LSTM and GRU models**, where I developed hybrid LSTM-GRU models for predicting the output voltage of various MFC's setups

SPARKLE Research Fellow

Oct 2023 - May 2025

Let's Rewire EV battery charging using ML

Rajasthan, India

- Received an **INR 2 Lakh grant** under the **SPARKLE Program** to pursue research and **model development**
- Led a **10-member team** to **execute experiments**, collect data, and develop the machine learning model

PRMO Scholar

Pre-Regional Mathematics Olympiad

- One of the 200 students from the Delhi Region who qualified for Regional Mathematics Olympiad (RMO)

Oct 2018

Delhi, India

EXTRA-CURRICULAR

Football - Rocketium

Won the Man of the Match Award for scoring a hat trick in the Cult Sports Fest, Bengaluru 2024.

Apr 2024